LigPlot3D: An Interactive Web-Based Framework for Geometric Analysis of Protein-Ligand Interactions

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***Abstract*—Molecular docking studies routinely generate multiple candidate binding poses, creating a practical need for rapid and interpretable inspection of protein–ligand interactions beyond numerical scoring functions. While established tools provide comprehensive interaction profiling or schematic visualization, they often require local installation or produce static outputs that limit interactive exploration. Here, this work presents *LigPlot3D*, a web-based application for interactive, client-side analysis of protein–ligand complexes. LigPlot3D combines browser-native three-dimensional molecular rendering with deterministic, geometry-based identification of key non-covalent interactions, including hydrogen bonds, salt bridges, hydrophobic contacts, π-stacking, and halogen bonds. All analysis is performed locally within the browser, ensuring data privacy and eliminating server-side dependencies. Using representative docking complexes, LigPlot3D enables rapid qualitative assessment of binding modes and interaction geometry through an intuitive interface that integrates interactive visualization with structured interaction tables. By emphasizing accessibility, interactivity, and reproducible geometric criteria, LigPlot3D complements existing interaction profilers and visualization tools and supports efficient post-docking analysis in early-stage structure-based workflows. LigPlot3D is freely available at https://singhnitink.github.io/LigPlot3D/, and its source code can be cloned from GitHub and run locally using npm for offline or customized use.**

***Index Terms*—Molecular Docking, Visualization, Web Application, Interaction Analysis.**

I. Introduction

Molecular docking remains a widely used computational approach for predicting plausible binding modes and approximate affinities of small molecules in protein binding sites [1], supporting structure-based drug discovery workflows and virtual screening pipelines [2]. In practice, docking campaigns typically generate multiple poses per ligand (and often multiple ligands per target), creating an immediate downstream need for rapid, reliable inspection of poses, ranking consistency, and the chemical plausibility of predicted binding modes. Beyond numerical scores, the interpretability of docking results often depends on verifying whether a pose reproduces key physicochemical determinants of binding, such as hydrogen-bond patterns and local contact geometry in the binding site.

A common strategy for increasing interpretability is the explicit identification of non-covalent interactions between ligand and receptor. Automated interaction profilers such as the Protein–Ligand Interaction Profiler (PLIP) provide systematic detection and reporting of interaction types in macromolecular structures, enabling consistent characterization of binding determinants across complexes [3]. Recent advances have extended PLIP to incorporate protein–protein interactions, reflecting the broader demand for scalable interface analysis as structural datasets and predicted complexes continue to grow [4]. These profilers offer robust interaction taxonomies and batch-oriented workflows, but they are not primarily designed as lightweight, web-first environments for rapid, interactive inspection of docking results across multiple poses and ligands.

Complementary to interaction profiling, schematic visualization tools such as LigPlot+ generate 2D interaction diagrams that summarize hydrogen bonds and non-bonded contacts, and support comparative analysis across related ligand–protein complexes [5]. While 2D schematics can be effective for communicating interaction hypotheses and for comparing congeneric series, docking analysis frequently benefits from interactive exploration of alternative poses, immediate cross-comparison of pose-specific interactions, and streamlined access without local installation—particularly in early-stage triage and iterative design settings.

To address these practical needs, this work presents *LigPlot3D*, a web-based application for interactive analysis of docking outputs and protein–ligand complexes. LigPlot3D is designed to support rapid pose-centric inspection by combining structure visualization with geometry-based identification of hydrogen bonds and contact-level summaries, enabling users to compare alternative poses and ligands within a unified, accessible interface. By emphasizing lightweight deployment and interactive workflows, LigPlot3D aims to complement established profilers and diagram generators by improving the usability of docking post-processing and facilitating reproducible, inspection-driven decision-making.

II. Model and Methods

A. Architectural Framework

LigPlot3D is implemented as a fully client-side Single Page Application (SPA), designed to prioritize interactive performance, data locality, and platform independence. All structure parsing, interaction analysis, and visualization are executed entirely within the user's web browser, eliminating server-side dependencies and preventing external transmission of structural data. In addition to the hosted web application, users can clone the source repository from GitHub and run LigPlot3D locally using standard npm commands (`npm install` followed by `npm run dev`), enabling offline use, custom modifications, and integration into local workflows.

The application is developed using **React.js** (v19.2) [6] with a functional component architecture. React hooks are employed for deterministic state management of molecular

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representations, interaction tables, and user-driven selections. The build and bundling pipeline is based on **Vite** (v7.2), which enables efficient module resolution and fast incremental rebuilds during development. Layout, typography, and responsive styling are handled using **TailwindCSS**, ensuring consistent rendering across display sizes and devices.

B. Molecular Visualization Engine

Three-dimensional molecular visualization is performed using a hybrid WebGL-based rendering layer that integrates **NGL Viewer** (v2.4.0) [7] and **3Dmol.js** (v2.5.3) [8]. NGL Viewer is used for rendering macromolecular structures, including protein cartoons, surfaces, and backbone representations, and provides robust support for standard structural formats such as PDB and mmCIF. In parallel, 3Dmol.js is employed for efficient rendering of small molecules and localized interaction geometries.

A custom synchronization layer coordinates camera state, orientation, and selection events between the two rendering engines, allowing seamless transitions between global protein views and ligand-focused inspection while preserving spatial context.

C. Geometric Interaction Algorithms

LigPlot3D incorporates a deterministic, geometry-based interaction detection module intended for rapid, pose-centric analysis of protein–ligand complexes. Interaction classification is performed using explicit distance- and angle-based criteria. Each criterion is adapted from a specific, previously published geometric convention for that interaction type, cited individually in the subsections below, rather than defined ad hoc for this work. All interaction calculations are executed client-side using atomic coordinates parsed directly from the loaded structure files.

Geometric thresholds and atom classifications are defined centrally within the application configuration (`constants.ts`), ensuring transparency and reproducibility of interaction assignments. All active thresholds are documented within the application interface, allowing users to inspect the criteria underlying interaction detection.

1) Hydrogen Bonds: Hydrogen bonds are identified between potential donor and acceptor atoms restricted to nitrogen, oxygen, and sulfur, following the heavy-atom donor–acceptor convention established for protein structures that lack resolved hydrogens [9]. A hydrogen bond is assigned when the donor–acceptor heavy-atom separation is less than or equal to 3.5 Å. When the donor's hydrogen position is resolvable from the loaded structure, LigPlot3D additionally enforces the donor–hydrogen–acceptor angle criterion of $120°$ or greater established by McDonald and Thornton [10]. Because most crystallographic and docking-derived structures omit explicit hydrogens, this angle check is applied conditionally, and the heavy-atom distance criterion alone governs assignment otherwise. Fluorine atoms are explicitly excluded from the acceptor set due to their limited hydrogen-bonding capacity in biological environments.

2) Salt Bridges: Salt bridges are defined as electrostatic interactions between oppositely charged ionizable groups. Positively charged centers include nitrogen atoms in the side chains of arginine (NH1, NH2), lysine (NZ), and histidine (ND1, NE2), while negatively charged centers correspond to carboxylate oxygen atoms in aspartate (OD1, OD2) and glutamate (OE1, OE2). A salt bridge is assigned when the distance between charge centers is less than or equal to 4.0 Å, adapted from the ion-pair distance convention reported by Barlow and Thornton [11].

3) Hydrophobic Contacts: Hydrophobic contacts are detected between non-polar carbon atoms, including both aliphatic and aromatic carbons. A distance cutoff of 4.5 Å is applied to capture van der Waals contacts that contribute to hydrophobic stabilization within the binding site, consistent with the hydrophobic contact ranges discussed for ligand–protein recognition by Bissantz, Kuhn, and Stahl [12].

4) π-Stacking Interactions: Aromatic interactions are evaluated for residues containing planar ring systems, including phenylalanine, tyrosine, tryptophan, and histidine. Ring centroids and plane normals are computed to classify interaction geometry, following the geometric classification of aromatic side-chain pairs reported by McGaughey, Gagné, and Rappé [13]. Interactions are assigned when the centroid–centroid distance is less than or equal to 5.5 Å. Parallel stacking interactions require an inter-planar angle of $30°$ or less with a lateral offset not exceeding 2.0 Å, while T-shaped stacking interactions are identified by inter-planar angles greater than or equal to $60°$. LigPlot3D also detects cation–π contacts between these aromatic ring centroids and positively charged ligand or protein groups within 5.0 Å, using the distance convention of Gallivan and Dougherty [14], and reports them under the π-stacking interaction class rather than as a separate category.

5) Halogen Bonds: Halogen bonds are identified between halogen donors (Cl, Br, I) and nucleophilic acceptors (N, O, S), following the directional geometric criteria for halogen bonding in biological molecules established by Auffinger and coworkers [15]. An interaction is assigned when the halogen–acceptor separation is less than or equal to 3.5 Å and the C–X$\cdots$acceptor angle, measured from the ligand carbon covalently bonded to the halogen, is greater than or equal to $140°$, consistent with sigma-hole directionality. Unlike the hydrogen-bond donor angle, this angle is always well-defined, because a halogen substituent is invariably bonded to exactly one heavy atom.

6) Metal Coordination: Metal coordination interactions are detected for common biologically relevant metals, including Zn, Fe, Mg, and Ca, interacting with electronegative atoms (N, O, S). A stringent distance threshold of 2.8 Å is applied to identify coordination contacts, within the range of metal–ligand coordination distances surveyed by Harding [16]. That survey shows optimal coordination distance varies by metal identity, so a single uniform cutoff across all supported metals is a simplification that trades per-metal precision for implementation simplicity. The current implementation relies exclusively on geometric criteria and does not incorporate energetic scoring or solvent models.

7) Applicability Beyond Protein Receptors: Hydrogen bond, halogen bond, hydrophobic contact, and metal coordination detection operate on generic element types rather than residue identity, so these interaction classes already apply to ligands bound to RNA or DNA receptors to a meaningful degree. Salt-bridge and receptor-side π-stacking detection remain restricted to the canonical amino-acid residue definitions in `constants.ts` and are not yet extended to nucleotide bases and backbone phosphate groups. Formalizing nucleic-acid-specific residue and ring definitions to bring these two interaction classes to parity is identified as a direct extension of this work.

III. Results and Discussion

A. Interactive Exploration of Binding Interfaces

LigPlot3D provides an integrated interface composed of a central three-dimensional visualization viewport and a configurable analytical sidebar. Upon loading a PDB structure, the application parses the coordinate file, identifies ligand entities, and constructs a localized representation of the binding environment. Users may selectively control protein representations, ligand styles, and interaction overlays through explicit interface controls.

The landing page allows users to drag-and-drop structure files directly into the browser window, as shown in Figure 1. Once a structure is loaded, LigPlot3D enables interactive inspection of key non-covalent interactions. An example interaction analysis for the JAK2 complex is shown in Figure 1. Users can dynamically toggle visibility for hydrogen bonds, salt bridges, hydrophobic contacts, π-stacking, and halogen bonds through the sidebar controls, allowing for focused analysis of specific interaction types.

B. Interaction Detection and Geometric Consistency

The interaction detection module was evaluated by examining representative protein–ligand complexes commonly used in docking benchmarks. For the Macrophage Migration Inhibitory Factor (MIF) complex [17], LigPlot3D identified canonical active site interactions consistent with structural reporting (Figure 1). The detected interactions demonstrated qualitative agreement with reference profiles, reflecting the deterministic nature of the geometry-based criteria.

The application applies strict distance and angular constraints for interaction assignment, which reduces the inclusion of weak or geometrically ambiguous contacts. In practice, this results in concise interaction maps that emphasize structurally meaningful interactions rather than exhaustive contact enumeration. While LigPlot3D does not aim to provide an exhaustive interaction taxonomy, the implemented interaction classes capture the dominant non-covalent features relevant for docking pose assessment.

C. Performance and Accessibility

All visualization and interaction analysis in LigPlot3D is executed client-side within a standard web browser. In routine use, the application loads typical protein–ligand complexes without perceptible delay and maintains smooth interactive frame rates for structures containing up to approximately 10^5 atoms. Because no server-side processing is required, LigPlot3D can be used in restricted or offline environments and does not transmit structural data externally.

This deployment model lowers the barrier to entry for docking result inspection and facilitates rapid review of binding hypotheses, making the tool suitable for early-stage pose triage, educational use, and collaborative discussion settings.

D. Comparison with Existing Tools

To contextualize LigPlot3D within the landscape of interaction analysis and visualization tools, its functionality was compared with LigPlot+ [5], the Protein–Ligand Interaction Profiler (PLIP) [3], [4], Arpeggio [18], ProLIF [19], and COCOMAPS [20], [21]. Table I summarizes deployment model and typical use case across these tools.

LigPlot+ generates publication-quality two-dimensional interaction schematics by projecting three-dimensional complexes into flattened diagrams. These representations are effective for static reporting and comparative summaries but inherently sacrifice spatial depth information. LigPlot3D differs in that it preserves the full three-dimensional geometry of the binding site, enabling users to interrogate interaction orientations and steric relationships interactively within the browser.

PLIP provides a comprehensive and well-validated interaction profiling framework, including interaction classes not yet implemented in LigPlot3D such as water bridges, and supports both protein–ligand and protein–protein interfaces. However, PLIP primarily operates through server-based reports or command-line workflows that generate static outputs or external visualization scripts. LigPlot3D complements this functionality by offering an interactive, client-side environment focused on rapid docking pose inspection and immediate visual feedback, without requiring data upload or local software installation. Table II compares LigPlot3D's geometric thresholds against PLIP's default configuration in detail.

Arpeggio calculates interatomic contacts directly from a structure file through an OpenBabel-based atom-typing pipeline, classifying a broad range of contact types, including ring interactions, hydrogen bonds, and steric clashes, through its web server and standalone library [18]. Its interaction taxonomy is more granular than LigPlot3D's, but Arpeggio is designed around single-structure, API-driven batch analysis rather than the pose-by-pose interactive comparison that LigPlot3D targets for docking triage.

ProLIF encodes protein–ligand and protein–protein interactions as bit-vector fingerprints using RDKit, and is built specifically for scoring and comparing large ensembles of poses or molecular dynamics frames [19]. This fingerprint representation is well suited to downstream machine learning and trajectory analysis, but ProLIF is used as a Python library within a scripting or notebook environment rather than as a standalone interactive viewer, so it does not provide LigPlot3D's direct visual inspection of a single pose.

COCOMAPS analyzes and visualizes residue-level contacts at the interface of biomolecular complexes, historically empha-

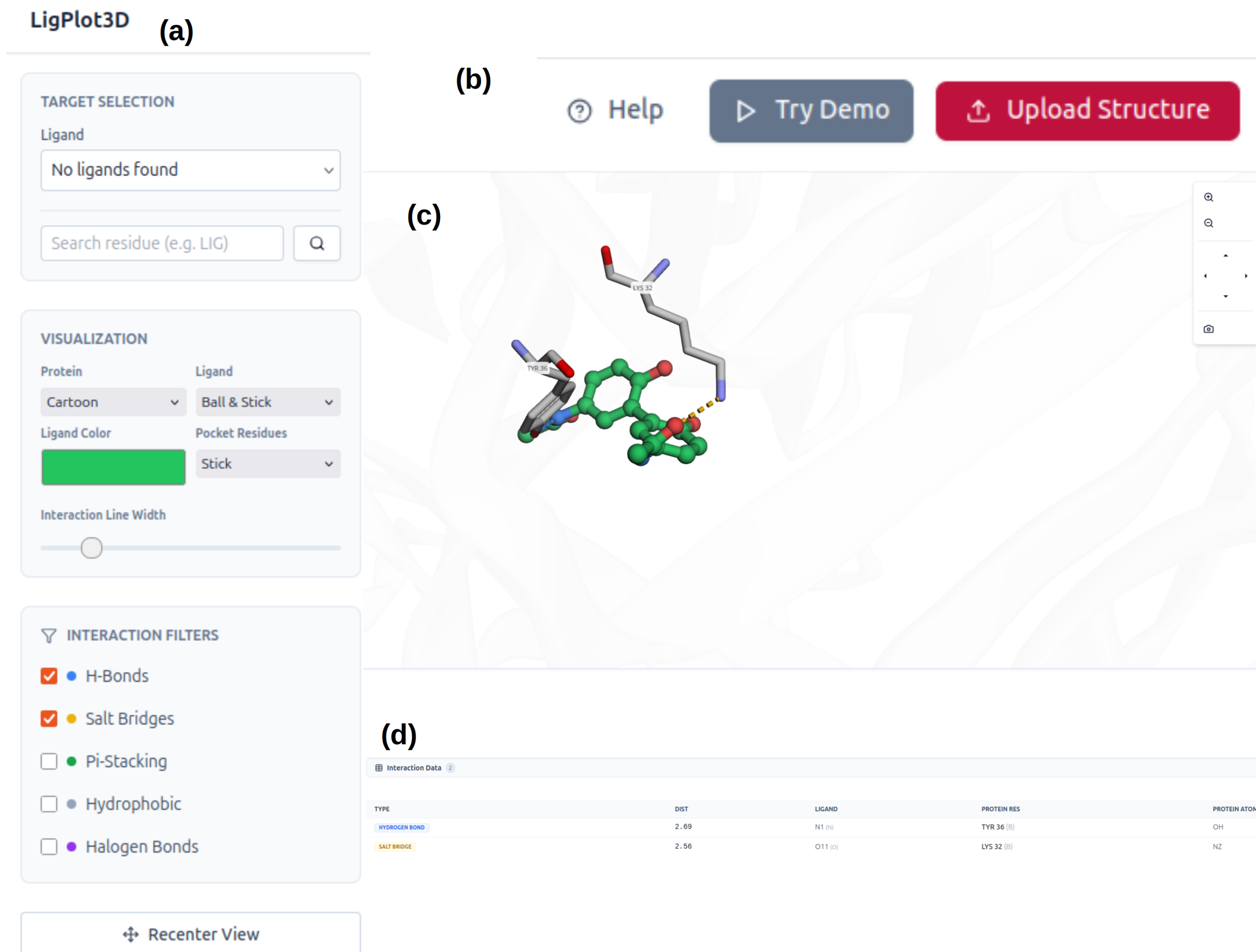

Fig. 1. User interface and workflow of the LigPlot3D web application. The clean and minimal interface of LigPlot3D is organized into four functional regions to streamline protein–ligand interaction analysis. (a) Control center: The left-hand panel provides target selection and visualization controls. (b) Structure input module: Users can upload a local PDB file or directly enter a PDB identifier. (c) Main visualization canvas: The central 3D rendering window displays the protein–ligand complex. (d) Analysis pane: A structured interaction table summarizes detected contacts.

sizing protein–protein and protein–nucleic acid interfaces over small-molecule ligands [20]. The recent COCOMAPS 2.0 release adds atomic-level interaction classification across sixteen contact types and interactive Mol* visualization [21], narrowing the visualization gap with LigPlot3D. COCOMAPS 2.0 remains a file-upload web-server workflow, however, so LigPlot3D differs by performing the equivalent client-side analysis locally, without transmitting structural data.

TABLE I
DEPLOYMENT MODEL AND TYPICAL USE CASE ACROSS INTERACTION ANALYSIS TOOLS.

Tool	Deployment	Interactive	Primary use case
LigPlot3D	Client-side web app	Yes	Interactive pose-by-pose docking triage
LigPlot+ [5]	Local desktop software	Limited	Publication-quality 2D interaction schematics
PLIP [3], [4]	Web server / CLI	No	Batch interaction profiling and reporting
Arpeggio [18]	Web server / library	No	Granular, API-driven contact calculation
ProLIF [19]	Python library	No	Interaction fingerprints for MD and pose ensembles
COCOMAPS 2.0 [21]	Web server	Yes (Mol*)	Interface contact-map analysis of biomolecular complexes

LigPlot3D, PLIP, Arpeggio, ProLIF, and COCOMAPS all classify interactions from geometric proxies rather than from computed interaction energies. Their outputs are therefore expected to agree closely on well-formed, unambiguous contacts, such as short and near-linear hydrogen bonds or face-to-face aromatic stacking, and to diverge most at marginal cases close to a distance or angle cutoff. Divergence at these margins reflects genuine differences in atom-typing granularity, in

TABLE II
LIGPLOT3D GEOMETRIC THRESHOLDS COMPARED WITH PLIP'S DEFAULT CONFIGURATION [22]. THE TWO TOOLS ARE INDEPENDENTLY PARAMETERIZED AND THEIR THRESHOLDS ARE NOT INTENDED TO BE IDENTICAL.

Interaction	LigPlot3D	PLIP (default)
Hydrogen bond distance	≤ 3.5 Å	≤ 4.1 Å
Hydrogen bond angle	$\geq 120°$ (D–H–A)^a	$\geq 100°$ (donor angle)
Salt bridge distance	≤ 4.0 Å	≤ 5.5 Å
Hydrophobic contact distance	≤ 4.5 Å	≤ 4.0 Å
π-stacking distance	≤ 5.5 Å	≤ 5.5 Å
π-stacking angle deviation	$\leq 30°$ (parallel)	$\leq 30°$
Halogen bond distance	≤ 3.5 Å	≤ 4.0 Å
Halogen bond angle	$\geq 140°$ (C–X$\cdots$A)	$165° \pm 30°$ (donor), $120°$ (acceptor)
Metal coordination distance	≤ 2.8 Å	≤ 3.0 Å

^{a}Applied when the donor's hydrogen is resolvable from the loaded structure. Otherwise, LigPlot3D applies the heavy-atom distance criterion alone (see the Hydrogen Bonds subsection).

whether donor hydrogens are placed or inferred, and in the specific threshold values each tool applies (Table II), rather than an error in any single implementation. None of these geometric criteria substitute for an energetic or experimental determination of interaction strength. Confirming that a geometrically detected contact is functionally significant, as opposed to a fortuitous short contact, requires orthogonal evidence such as mutagenesis, isothermal titration calorimetry, or quantum-mechanical interaction energy calculations. A systematic, structure-matched benchmark of LigPlot3D against PLIP, Arpeggio, ProLIF, and COCOMAPS, together with comparison against such experimentally validated interaction sets, is identified as a direct extension of this work.

IV. Conclusion

In this study, LigPlot3D is presented as a web-based application for interactive analysis of protein–ligand binding interfaces. LigPlot3D integrates browser-native three-dimensional molecular visualization with deterministic, geometry-based detection of key non-covalent interactions, enabling rapid qualitative assessment of docking poses and binding modes.

By performing all parsing, visualization, and interaction analysis client-side, LigPlot3D eliminates the need for local software installation or remote server execution, while preserving data privacy. The application is designed to complement established interaction profilers and schematic visualization tools by emphasizing interactivity, accessibility, and immediate structural feedback within a lightweight workflow.

LigPlot3D is particularly well suited for early-stage docking analysis, educational use, and rapid inspection of binding hypotheses where intuitive exploration of interaction geometry is essential. Future development will focus on extending supported interaction annotations, incorporating optional energy-related descriptors, formalizing support for RNA and DNA receptors, and enabling analysis of time-resolved structural data such as molecular dynamics trajectories.

Data and Software Availability

LigPlot3D is open-source software released under the MIT License.

- **Web Application:** https://singhnitink.github.io/LigPlot3D/
- **Source Code:** https://github.com/singhnitink/LigPlot3D

Conflicts of Interest

The author declares no conflict of interest.

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